

THESIS PROPOSAL

**24-Hour Ahead PM_{2.5} Concentration Forecasting in Dhaka
Using Historical Air Quality and Meteorological Data**

Abstract

Air pollution, particularly fine particulate matter (PM_{2.5}), is one of the most critical environmental and public health challenges in Dhaka, Bangladesh. Dhaka regularly ranks among the world's most polluted megacities, with PM_{2.5} concentrations frequently exceeding the WHO 24-hour guideline of 15 µg/m³ by factors of 5 to 10 during winter months. Despite the presence of continuous real-time monitoring infrastructure, the absence of a reliable short-term forecasting system severely limits the capacity of government agencies, healthcare institutions, schools, and citizens to take proactive protective action before hazardous air quality events occur.

This thesis proposes a 24-hour ahead PM_{2.5} forecasting framework for Dhaka, leveraging a multivariate time-series dataset of hourly air quality and meteorological observations collected from March 2016 to June 2022. The proposed approach will integrate advanced feature engineering, leakage-safe data preprocessing, Optuna-based hyperparameter optimization, and a multi-model ensemble combining Gradient Boosting (LightGBM, XGBoost, CatBoost), Deep Learning (CNN-BiLSTM with Bahdanau Attention), and classical statistical baselines.

The primary forecasting target is the 24-hour-ahead daily-average PM_{2.5} concentration — the standard AQI/EPA/WHO bulletin forecasting definition — which is both practically actionable for public health advisories and more tractable than point-specific hourly prediction. The proposed framework is designed to operate using only standard single-station monitoring and meteorological inputs, making it deployable in resource-constrained environments characteristic of developing nations.

1. Introduction

1.1 Background and Motivation

Particulate Matter with aerodynamic diameter ≤ 2.5 micrometres (PM_{2.5}) is classified by the World Health Organization as a Group 1 carcinogen with no demonstrated safe lower threshold of exposure. Its fine size enables deep penetration into the respiratory tract and bloodstream, where it contributes to cardiovascular disease, pulmonary disorders, stroke, and premature mortality. According to the WHO Global Air Quality Guidelines (2021), the annual mean PM_{2.5} guideline is 5 $\mu\text{g}/\text{m}^3$ and the 24-hour mean guideline is 15 $\mu\text{g}/\text{m}^3$.

Dhaka, the capital of Bangladesh and one of the world's most densely populated megacities with over 20 million residents, consistently ranks among the five most polluted cities globally. The city's PM_{2.5} concentrations exhibit extreme seasonal variability driven by its geographic and climatic conditions: winter months (December through February) frequently record hourly means exceeding 150 to 200 $\mu\text{g}/\text{m}^3$, driven by atmospheric temperature inversions, reduced planetary boundary layer height, fog events, and biomass burning; while monsoon months (June through August) provide relative relief with concentrations falling to 30 to 40 $\mu\text{g}/\text{m}^3$ through rainfall washout and strengthened southwest winds.

Despite the presence of continuous PM_{2.5} monitoring through the US Embassy AirNow station, the current monitoring infrastructure operates entirely in a reactive mode — describing the present state of air quality with no capacity to predict future pollution levels. Government agencies, hospital systems, schools, and industrial operators all require at least 12 to 24 hours of lead time to implement protective measures effectively. The development of a reliable 24-hour forecasting system therefore addresses a critical operational gap in Dhaka's public health infrastructure.

1.2 Problem Statement

The core problem addressed by this thesis is: given historical hourly observations of PM_{2.5} concentration and concurrent meteorological variables — temperature, relative humidity, wind speed, and rainfall — recorded up to time T , predict the average PM_{2.5} concentration over the 24-hour window spanning from $T+1$ to $T+24$ hours.

This forecasting task is characterized by three fundamental challenges that distinguish it from simpler regression problems. First, PM_{2.5} in Dhaka exhibits high temporal variability and non-stationarity, with concentrations driven by a complex mixture of meteorological forcing, seasonal emission patterns, and sporadic event-driven spikes from fires, construction, festivals, and traffic congestion. Second, extreme pollution spikes — events where PM_{2.5} rises rapidly by 100 $\mu\text{g}/\text{m}^3$ or more within a few hours — are driven by localized factors not captured by standard meteorological observations alone, making them intrinsically difficult to forecast from available features. Third, the 24-hour forecast horizon is substantially harder than shorter horizons: autocorrelation in PM_{2.5} weakens considerably between the 1-hour and 24-hour lag, meaning that the predictive signal available from recent observations degrades substantially over the forecast window.

1.3 Research Objectives

This thesis pursues the following specific research objectives:

- To design and implement a comprehensive data preprocessing pipeline that correctly handles sensor errors, outliers, temporal gaps, and the strict chronological ordering requirements of time-series data — ensuring no future information leaks into training features.
- To develop an advanced feature engineering framework capturing the dominant signal sources in Dhaka's PM_{2.5} data: lag autocorrelation at multiple scales, exponential moving

averages, rolling statistical summaries, meteorological lags, and physics-motivated interaction features.

- To train and evaluate multiple model architectures spanning classical statistical baselines (Persistence, SARIMA), state-of-the-art gradient boosting models (LightGBM, XGBoost, CatBoost) with Optuna-based hyperparameter optimization, and a novel deep learning architecture (CNN-BiLSTM with Bahdanau Attention) designed to capture both local spike patterns and long-range weekly seasonality.
- To construct an optimised weighted ensemble that selectively combines the best-performing base models using validation-set performance, and to compare its accuracy against each individual model and published benchmarks.
- To conduct a SHAP-based feature attribution analysis identifying which meteorological and temporal features are most predictive of next-day PM2.5 levels, providing scientifically interpretable insights into Dhaka's pollution dynamics.
- To analyse forecast accuracy across multiple prediction horizons (1h, 3h, 6h, 12h, 24h), demonstrating the expected degradation pattern and situating the 24h result within the broader time-series forecasting literature.
- To compare achieved results against the best published comparable studies in similarly volatile, high-pollution urban environments, with particular reference to the only prior Dhaka-specific PM2.5 forecasting studies.

1.4 Significance of the Study

This research makes three distinct contributions. First, it provides the most comprehensive PM2.5 forecasting study specifically targeting Dhaka, Bangladesh — a city whose pollution dynamics (monsoon-driven seasonality, informal industrial activity, riverine topography, and dense informal transport) differ substantially from the Chinese and European cities that dominate the existing ML forecasting literature. Second, the proposed framework is designed to operate using only variables available from a single monitoring station and standard meteorological records, requiring no satellite aerosol optical depth data, chemical speciation sensors, or traffic density measurements — a practical requirement for deployment across the resource-constrained monitoring networks found in developing nations. Third, the multi-model comparative study with Optuna optimization and SHAP interpretability provides a rigorous, reproducible benchmark that future researchers studying Dhaka or comparable South Asian urban environments can directly extend.

2. Literature Review

2.1 Overview of PM2.5 Forecasting Methods

The field of data-driven PM2.5 forecasting has evolved through three broad generations of approaches. Classical statistical models such as ARIMA and SARIMA established the baseline for univariate time-series forecasting of pollutant concentrations, offering interpretability and computational efficiency but limited capacity to model nonlinear relationships or incorporate multivariate inputs. The second generation of machine learning approaches — Random Forest, Support Vector Regression, Gradient Boosting — introduced the ability to exploit complex nonlinear relationships between PM2.5 and meteorological predictors, yielding significant accuracy improvements over statistical baselines. The current generation of deep learning architectures — LSTM, CNN-LSTM, Transformer, and hybrid attention-based models — further extends the ability to capture long-range temporal dependencies and spatial patterns across monitoring networks.

A key finding from the literature is that reported R^2 values vary substantially with forecast horizon and target definition. Studies achieving R^2 above 0.95 almost universally use either next-hour prediction, daily-average data rather than hourly measurements, additional satellite or chemical speciation inputs, or cities with substantially lower PM2.5 variance than Dhaka. The 24-hour ahead forecasting task in high-pollution, meteorologically-driven environments consistently yields R^2 in the range of 0.65 to 0.85 across all major published studies.

2.2 Most Relevant Published Studies

The following studies are most directly relevant to this thesis, organized by their proximity to the specific task (24h-ahead, meteorological features only, high-pollution city):

Study	Location	Model	R^2 /RMSE	Relevance to This Thesis
Islam et al. (2021) — Atmosphere	Dhaka, Bangladesh	CatBoost, ARIMA-ANN, ARIMA-SVM	~0.78 (CatBoost best)	Only prior ML study specifically targeting Dhaka PM2.5 — direct benchmark
Zaman et al. (2023) — Air Quality, Atmosphere & Health (Springer)	Dhaka, Bangladesh	Best Subset Regression + ML	~0.75	Same city, meteorological features — second Dhaka benchmark
Mohammadi et al. (2025) — ISPRS Int. J. Geo-Information	Tehran, Iran	3D CNN-GRU with DTW	$R^2=0.84$, RMSE=7.15 $\mu\text{g}/\text{m}^3$	Most comparable study in a volatile high-pollution city; 24h ahead horizon
Chen et al. (2024) — PeerJ	China	CNN-LSTM	$R^2=0.91$, RMSE=8.22 $\mu\text{g}/\text{m}^3$	Useful architecture reference; daily prediction, lower-variance city
Liu et al. (2023) — ScienceDirect	Wuhan, China	SA-LSTM + DTW	$R^2=0.95$, RMSE=6.92 $\mu\text{g}/\text{m}^3$	Daily average data — easier task; EMA/DTW feature ideas adopted
PMC Dual-Channel CNN (2025)	Shanghai/Taiwan	Dual-Channel CNN (surveillance + meteo)	$R^2=0.94\text{--}0.97$	Not comparable — uses camera image inputs not available in this study
PMC Transformer-LSTM (2025)	Chengdu, China	Transformer-LSTM	>0.90	Seasonal/monthly horizons — not strict 24h; architecture inspiration

2.3 Key Observations from Literature

- High R^2 values (>0.93) reported in the literature are almost exclusively from studies using daily-average data, extra satellite inputs, or cities with substantially lower PM2.5 variance than Dhaka (mean $88 \mu\text{g}/\text{m}^3$, std $77 \mu\text{g}/\text{m}^3$, skewness 1.61).
- The only two published studies targeting Dhaka specifically (Islam et al., 2021; Zaman et al., 2023) achieved R^2 of approximately 0.75–0.78, using simpler model architectures and smaller datasets. This thesis directly extends those works.
- The Tehran 3D CNN-GRU study (Mohammadi et al., 2025) is the most comparable in terms of task difficulty (volatile, high-pollution, meteorological features only, 24h horizon), reporting $R^2=0.84$. Matching or exceeding this benchmark represents a primary target.
- Gradient boosting models consistently match or outperform deep learning approaches on structured tabular time-series features when feature engineering is thorough, particularly when training data is limited to a single monitoring station.
- MAE (L1) loss functions outperform MSE for right-skewed PM2.5 distributions because they reduce the disproportionate influence of extreme spike events on parameter estimation.

3. Proposed Methodology

3.1 Dataset Description

The primary dataset is constructed by merging two sources: (1) hourly PM2.5 measurements from the US Embassy AirNow monitoring station in Dhaka, collected across seven annual CSV files covering January 2016 to June 2022, sourced from the US Department of State's Air Quality Monitoring Program; and (2) a matched hourly meteorological dataset providing temperature ($^{\circ}\text{C}$), relative humidity (%), wind speed (km/h), and rainfall (mm) for the same period. After merging and initial cleaning, the dataset contains 51,914 hourly records spanning March 2016 to June 2022.

Parameter	Details
Study period	March 2016 – June 2022 (6.3 years)
Temporal resolution	Hourly
Total merged records	51,914
PM2.5 source	US Embassy AirNow station, Dhaka
Meteorological variables	Temperature, Relative Humidity, Wind Speed, Rainfall
PM2.5 overall mean	$88.3 \mu\text{g}/\text{m}^3$
PM2.5 standard deviation	$77.5 \mu\text{g}/\text{m}^3$
PM2.5 maximum observed	$438.4 \mu\text{g}/\text{m}^3$
Distribution skewness	1.61 (right-skewed heavy tail)
Winter mean (Dec–Feb)	$\sim 168 \mu\text{g}/\text{m}^3$ — Hazardous
Monsoon mean (Jun–Aug)	$\sim 34 \mu\text{g}/\text{m}^3$ — Moderate

3.2 Data Preprocessing Plan

The preprocessing pipeline will apply the following steps in sequence, each with explicit documentation of the rationale and magnitude of each correction:

- Removal of sensor-error readings ($\text{PM}_{2.5} < 1 \mu\text{g}/\text{m}^3$), which represent physically implausible measurements in Dhaka's ambient environment.

- Winsorization of extreme outliers — values above the 99.5th percentile will be capped rather than removed, preserving temporal continuity for rolling-window feature calculations while limiting the influence of isolated extreme readings on loss function gradients.
- Merging of annual PM2.5 files with the meteorological dataset on the datetime field, after standardising all datetime formats to UTC+6 (Bangladesh Standard Time).
- Reindexing to a continuous hourly grid to expose temporal gaps explicitly. Small gaps of up to 6 consecutive hours will be filled using time-linear interpolation; larger gaps will be retained as NaN to prevent lag features from silently spanning data voids.
- Explicit leakage verification: every feature column will be confirmed to reference only data from time $\leq T$ before the target column is computed.

3.3 Feature Engineering Plan

A comprehensive feature engineering pipeline will produce approximately 60 input features from the 6 raw variables. Feature categories will include:

- PM2.5 lag features at 16 time steps: 1, 2, 3, 4, 5, 6, 8, 12, 18, 24, 36, 48, 72, 120, 168, and 336 hours — capturing autocorrelation structure at all key lags identified through ACF analysis.
- Exponential Moving Averages (EMA) at spans of 6, 12, 24, 48, and 72 hours — EMA weights recent observations more heavily than simple rolling averages, making it better suited for trending pollution episodes.
- Rolling statistics (mean, standard deviation, maximum, minimum, median, Q75, Q25, IQR) at windows of 6, 12, 24, 48, 72, and 168 hours.
- Momentum and rate-of-change features: first differences at 1h, 3h, and 24h; second difference (acceleration).
- Meteorological lag features: temperature, humidity, wind speed, and rainfall shifted by 1, 6, 12, 24, and 48 hours; 24-hour rolling means and standard deviations.
- Calendar features: hour, month, day of week, season (4 Dhaka-specific seasons), binary flags for rush hour, weekend, and nighttime, plus cyclical sine/cosine encodings for hour, month, and day of year.
- Interaction features: wind–humidity product, stagnation index (humidity / (wind speed + 1)), cumulative rainfall sums, calm-wind and high-pollution binary flags.

Feature selection will reduce the initial candidates to the final set using three sequential filters: variance threshold, high-collinearity removal ($|r| > 0.97$) favouring the higher-MI member, and mutual information threshold (MI < 0.05 bits excluded).

3.4 Target Variable Definition

The forecast target will be defined as: $\text{target}(t) = \text{mean}(\text{PM2.5}[t+1, t+2, \dots, t+24])$ — the arithmetic mean of PM2.5 over the 24-hour window immediately following the current observation. This definition aligns with the US EPA AQI-bulletin standard and the WHO daily-average air quality reporting metric. It is scientifically preferable to single-hour point prediction for the 24h horizon because the sub-daily within-day PM2.5 variability (estimated at approximately $\pm 34 \mu\text{g}/\text{m}^3$ average swing in Dhaka) represents an irreducible noise component that no available feature set can resolve 24 hours in advance — this noise is partially averaged out in the daily-mean target, producing a genuinely more predictable quantity.

3.5 Data Splitting Strategy

The dataset will be divided into training (70%), validation (15%), and test (15%) partitions. A block-interleaved strategy will be used: the timeline is first divided into contiguous 2-week blocks, which are then randomly assigned across splits using a fixed random seed. This approach addresses

a systematic distribution shift problem identified during preliminary data analysis — the later study years (2021–2022) have systematically higher mean PM2.5 than the earlier years (2016–2020), meaning a strict chronological split assigns disproportionately high-pollution periods to the test set only, producing an artificial train-test distributional mismatch. The block-interleaved split ensures each partition contains samples from all years and all seasons, producing comparable target distributions across splits. All preprocessing statistics (medians, scalers) will be computed exclusively on the training partition.

3.6 Proposed Model Architectures

3.6.1 Baseline Models

Two baselines will be established: (1) Persistence model — predicts the next-day average PM2.5 as equal to the current 24-hour rolling mean, serving as the minimum acceptable accuracy threshold; (2) SARIMA — a seasonal auto-regressive model representing the best achievable classical statistical forecast.

3.6.2 LightGBM

LightGBM will be trained using the MAE (regression_l1) objective, which is more appropriate than MSE for the right-skewed PM2.5 distribution. Hyperparameters (n_estimators, learning_rate, num_leaves, max_depth, min_child_samples, subsample, colsample_bytree, reg_alpha, reg_lambda) will be optimized using Optuna with 80 trials and the TPE sampler. The final model will be retrained on the combined train+validation data.

3.6.3 XGBoost

XGBoost will be trained using the reg:absoluteerror (MAE) objective with GPU acceleration (tree_method='hist', device='cuda'). Optuna hyperparameter optimization with 60 trials will cover the same parameter space as LightGBM. GPU acceleration is expected to reduce trial runtime substantially, enabling a more thorough search within the computational budget.

3.6.4 CatBoost

CatBoost will be trained using the MAE loss function with Optuna optimization over 50 trials. CatBoost's symmetric tree structure and built-in handling of categorical features make it a useful complement to LightGBM and XGBoost's leaf-wise and depth-wise strategies, potentially capturing different aspects of the feature space.

3.6.5 CNN-BiLSTM with Bahdanau Attention

The proposed deep learning architecture will process sequences of the preceding 168 hours (1 week) of the selected features. The architecture consists of: two causal 1D-Convolutional layers for local temporal pattern extraction (specifically designed to detect the sharp hour-to-hour transitions associated with pollution spike onset); Batch Normalization and Dropout for regularization; two Bidirectional LSTM layers for modelling long-range dependencies in both temporal directions across the full week context window; a Bahdanau Attention mechanism that computes scalar attention weights over each time step, enabling the model to focus on the most predictively relevant hours rather than treating all hours equally; and a dense output head. All hyperparameters (LSTM units, CNN filters, dropout rate, learning rate, batch size, dense units) will be selected via a 25-trial Optuna search.

3.6.6 Optimised Weighted Ensemble

The final prediction will be produced by a weighted ensemble of the base models. Ensemble weights will be optimized using the Scipy Nelder-Mead algorithm to minimize validation-set MAE, constrained to non-negative weights summing to 1.0. Only models whose validation R² exceeds 0.60 will be included in the ensemble — this threshold prevents weak learners from

degrading overall ensemble performance. The ensemble will be compared against both the best individual model and a simple average baseline.

3.7 Evaluation Metrics

Model performance will be assessed using four complementary metrics: (1) Coefficient of determination R^2 — measures the proportion of target variance explained; (2) Root Mean Squared Error (RMSE) in $\mu\text{g}/\text{m}^3$ — penalises large prediction errors more than small ones; (3) Mean Absolute Error (MAE) in $\mu\text{g}/\text{m}^3$ — robust to the heavy-tailed PM2.5 distribution; (4) Mean Absolute Percentage Error (MAPE) — provides a relative error measure useful for cross-study comparison. Performance will also be reported separately across five WHO/AQI pollution severity bands (Good 0–50, Moderate 50–100, Unhealthy for Sensitive Groups 100–150, Unhealthy 150–200, Very Unhealthy 200+) to characterise model behaviour across the full concentration range.

3.8 SHAP Interpretability

SHAP (SHapley Additive exPlanations) TreeExplainer analysis will be applied to the best-performing gradient boosting model to quantify the contribution of each feature to each prediction. SHAP summary plots and bar plots will be produced for inclusion in the thesis, enabling interpretation of which meteorological and temporal features drive next-day PM2.5 predictions — providing domain-relevant insights into Dhaka's pollution dynamics beyond pure predictive accuracy.

4. Research Timeline

Phase	Activity	Duration	Target Completion
Phase 1	Problem definition, literature review, dataset collection	2 months	Month 2
Phase 2	Data cleaning, merging, EDA, quality audit	1 month	Month 3
Phase 3	Feature engineering, ACF analysis, feature selection	1 month	Month 4
Phase 4	Baseline models and gradient boosting (LightGBM, XGBoost, CatBoost) with Optuna	2 months	Month 6
Phase 5	CNN-BiLSTM+Attention architecture and training	1 month	Month 7
Phase 6	Ensemble construction, SHAP analysis, evaluation	1 month	Month 8
Phase 7	Thesis writing, figures, tables, documentation	2 months	Month 10
Phase 8	Supervisor review, revision, defense preparation	1 month	Month 11

5. Tools and Technologies

Category	Tool / Library	Purpose
Programming	Python 3.12	Core development language
Data Processing	Pandas, NumPy	Data manipulation and numerical computation
ML — Gradient Boosting	LightGBM, XGBoost, CatBoost	Primary model architectures

Category	Tool / Library	Purpose
ML — Deep Learning	TensorFlow 2.x / Keras	CNN-BiLSTM+Attention implementation
Hyperparameter Opt.	Optuna (TPE sampler)	Bayesian hyperparameter optimization
Ensemble Optimization	SciPy (Nelder-Mead)	Ensemble weight search
Feature Selection	scikit-learn	Variance threshold, MI calculation
Interpretability	SHAP TreeExplainer	Feature attribution analysis
Compute	Kaggle Notebooks (2× Tesla T4 GPU)	Model training
Visualization	Matplotlib, Seaborn	EDA and result figures
Version Control	Git / GitHub	Code management

6. Expected Outcomes

On successful completion of this research, the following outcomes are anticipated:

- A validated, end-to-end PM2.5 forecasting pipeline for Dhaka that is fully documented and reproducible on the Kaggle platform.
- A rigorous comparative evaluation of six model architectures (Persistence, SARIMA, LightGBM, XGBoost, CatBoost, CNN-BiLSTM+Attention, and an optimised ensemble) on a held-out test set.
- Model accuracy meeting or exceeding the Tehran 3D CNN-GRU benchmark ($R^2=0.84$, Mohammadi et al., 2025) — the best published result for a comparable task in a similarly volatile, high-pollution urban environment.
- A SHAP interpretability analysis identifying the dominant predictors of next-day PM2.5 in Dhaka, contributing domain-relevant scientific insights alongside predictive accuracy.
- A multi-horizon analysis demonstrating forecast accuracy at 1h, 3h, 6h, 12h, and 24h horizons, contextualising the 24h result within the established literature pattern of accuracy degradation with forecast horizon.
- A published or submitted journal paper documenting the methodology and results, targeting an air quality, environmental informatics, or applied ML venue.

7. References

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